

CS208 Cloud Computing and Cloud Networking

Assignment 5: Container Orchestration

Kubernetes

Kubernetes is an open-source orchestration system for automating deployment, scaling, and management of containerized applications. You can think of it as OpenStack for Containers.

Problem Statement

In this assignment, you have to create two replicas of Nginx server pods using the Kubernetes Deployment object. After getting two replicas, create a Kubernetes Service to get a Virtual IP address and show that you are able to access the container using Virtual IP. You can use any command-line tool like curl, loadtest, hey, vegeta to access the Nginx service.

General Instructions

- You have to set up a multi node Kubernetes cluster. Use the small-lan profile with 2 nodes on Cloudlab, one acting as master and the other as a worker node.
- You have to create 2 Nginx pod replicas.
- You have to create the appropriate IP for the service you have deployed.
- Please follow the instructions for the installation steps given in the resources below.

Resources

- Kubernetes Installation: <https://www.mirantis.com/blog/how-install-kubernetes-kubeadm/>
- Nginx pods installation: <https://medium.com/@mngaonkar/kubernetes-get-started-deploy-a-simple-web-server-9636f4aa8706>
- Service: <https://kubernetesbyexample.com/services/>

Grading

What is tested?

- Screenshot of master node displaying the number of nodes connected to it. You should use `kubectl get nodes` to get this information.
- deployment.yaml file for creating Nginx pods with 2 replicas
- service.yaml file for creating service IP for the deployment.

Grade Distribution

- Screenshot of setting up Kubernetes cluster: 30 points
- Deployment.yaml file: 35 points
- Service.yaml file: 35 points